Systems & Control Letters 10 (1988) 9–16
North-Holland

Adaptive computed torque control for rigid link manipulations

R.H. MIDDLETON and G.C. GOODWIN

Department of Electrical and Computer Engineering, University of Newcastle, New South Wales, 2308, Australia

Received 26 June 1987
Revised 15 September 1987

Abstract: In this paper we shall examine the adaptive control of rigid link manipulator systems. Linear estimation techniques together with a computed torque control law are shown to give a globally convergent adaptive system which does not require measurements of accelerations.

Keywords: Adaptive control, Robotics, Non-linear systems, Computed torque control.

1. Introduction

The desire for improved dynamic accuracy in mechanical manipulators has led to research into improved control systems. Early works include the computed torque method of control (e.g. Paul [13] or Markewicz [11]) and adaptive controllers which treat the robot as a time varying linear system (e.g. Koivo and Guo [9], Dubowsky and DesForges [4], Tomizuka and Horowitz [18] or Koivo and Sorvari [10]). A complete analysis for this latter case is not available at present, however.

Hyperstability theory (e.g. Nicosia and Tomei [12]), sliding mode theory (e.g. Sastry and Slotine [15]), and stable factorization controllers (e.g. Spong and Vidyasagar [17]) have been used to propose adaptive control laws. However, these approaches suffer from one or more of the following difficulties: (i) non-zero tracking errors, (ii) unbounded feedback gains or (iii) 'chattering' effects. Recent work by Singh [16] has overcome the above difficulties, in the case of a revolute joint robot

In this paper we use a robot model which is linear in the unknown parameters. Parameter estimation techniques for robotic systems which require measurement of additional torques or accelerations, have been proposed in many references, for example Khasla and Kanade [7] or Chae, Atkeson and Hollerback [2]. This paper has its origins in the work of Elliott et al. [5], where a discrete time algorithm is considered. Craig, Hsu and Sastry [3] have proposed a linear parameterization, based on acceleration measurement, together with an adaptive computed torque control law and have established global convergence for their scheme. Bastin, Campion and Guillaume [1] have examined a similar scheme, and its robustness to bounded disturbances. Here we extend these works in the following areas: (i) we give a decoupled estimation and control procedure, (ii) measurements of extra torques or accelerations is not required and (iii) least squares estimation is given.

2. Linear parameter estimation for robots

The dynamic equations of motion for a general rigid link manipulator having n degrees of freedom can be described (using Lagrangian mechanics, eg. Kibble [8]) as follows:

$$\frac{\mathrm{d}}{\mathrm{d}t}\{I(\theta)\dot{\theta}\} = \tau_G(\theta) + \tau_M + \frac{1}{2}\frac{\partial}{\partial\theta}\left[\dot{\theta}^{\mathrm{T}} I(\theta)\dot{\theta}\right] \tag{1}$$

where θ, $\dot{\theta} \in \mathbb{R}^n$ are the robot angles, and angular velocities.

0167-6911/88/$3.50 © 1988, Elsevier Science Publishers B.V. (North-Holland)

$I(\theta) = I^{\mathrm{T}}(\theta) > 0$, $I(\theta) \in \mathbb{R}^{n\times n}$ is the inertia matrix and τ_M, $\tau_G(\theta) \in \mathbb{R}^n$ are the gravity torques, and actuator torques. We wish to perform on-line identification, since some parameters, particularly load related parameters, can undergo changes as the robot performs different tasks. It is well known (e.g. [2]) that the dynamic equations (1) may be written in a form that is linear in the unknown parameters such as mass and moments of interia. This can also be ext nded to include estimation of effects such as non-linear friction (see [3]). We now define:

$$I(\theta) = J(\theta) + \sum_{k=1}^{N} K_k(\theta) m_k \tag{2}$$

where $J(\theta)$ represents the known portion of the inertia matrix, $K_k(\theta)$ is a known function of θ, and m_k, $k = 1, \ldots, N$, are the unknown parameters. Clearly we can also write

$$\tau_G(\theta) = G(\theta) + \sum_{k=1}^{N} H_k(\theta) m_k \tag{3}$$

and in view of (1) we shall write

$$\frac{1}{2}\frac{\partial}{\partial\theta}\left[\dot\theta^{\mathrm{T}} I(\theta)\dot\theta\right] = D(\theta, \dot\theta) + \sum_{k=1}^{N} E_k(\theta, \dot\theta) m_k. \tag{4}$$

Using (2)–(4) we can rewrite (1) as

$$\phi^{\mathrm{T}} M = Y + \tau_M \tag{5}$$

where

$$Y = -\frac{\mathrm{d}}{\mathrm{d}t}\{J(\theta)\dot\theta\} + G(\theta) + D(\theta, \dot\theta)\}, \tag{6}$$

$$M^{\mathrm{T}} = [m_1, m_2, \ldots, m_N], \tag{7}$$

$\phi \in \mathbb{R}^{N\times n}$, and the k-th row of ϕ is

$$\phi_k = \left[\frac{\mathrm{d}}{\mathrm{d}t}\{K_k(\theta)\dot\theta\} - H_k(\theta) - E_k(\theta, \dot\theta)\right]^{\mathrm{T}}. \tag{8}$$

Note that equation (5) is linear in the parameter vector M and so linear estimation can be used provided ϕ, Y and τ_m are available. Note, however, that both ϕ and Y involve $\ddot\theta$. In this paper we shall show how estimation may be performed based on state variable measurements only. To achieve this, we operate on the left of (5) by $w/(D+w)$ where $D = \mathrm{d}/\mathrm{d}t$ and w is some constant:

$$\Psi^{\mathrm{T}} M = Y_F + \tau_F \tag{9}$$

where

$$\Psi = \left(\frac{w}{D+w}\right)\phi, \tag{10}$$

$$Y_F = \left(\frac{w}{D+w}\right) Y, \tag{11}$$

and

$$\tau_F = \left(\frac{w}{D+w}\right)\tau_m. \tag{12}$$

Ψ and Y_F are functions of θ and $\dot\theta$, but not $\ddot\theta$. Given an estimate $\hat M$ of M, define the vector prediction error $e(\hat M)$ by

$$e(\hat M) = Y_F + \tau_F - \Psi^{\mathrm{T}}\hat M = -\psi^{\mathrm{T}}\tilde M \tag{13}$$

where

$$\tilde{M} \triangleq \hat{M} - M. \tag{14}$$

Parameter estimation procedures can be derived based on the above equations by simple extensions to the estimators in (for example) Goodwin and Mayne [6]. We propose the following unnormalized least squares estimation algorithm:

$$\dot{\hat{M}} = P\Psi e, \tag{15}$$

$$\dot{P} = -P\Psi\Psi^{\mathrm{T}}P, \tag{16}$$

where $P \in \mathbb{R}^{N\times N}$, $P(0) = P(0)^{\mathrm{T}} > 0$. The following lemma establishes some of the properties of the above estimator.

Lemma 1. *The estimator*, (15), (16), *applies to the system* (9) *yields the following properties, regardless of the control law*:
(i) $\hat{M}$ *is bounded, and*
(ii) e *belongs to* L_2.

Proof. Consider the function

$$V = \tilde{M}^{\mathrm{T}}P^{-1}\tilde{M}. \tag{17}$$

Then using (13)–(16) we can show that

$$\dot{V} = -e^{\mathrm{T}}e. \tag{18}$$

The desired results then follow. □

Note also that any positive semi-definite modification to (16) can be included provided P remains bounded for all time.

3. Computed torque servo control

Several authors (see for example Sahba and Mayne [14], Markewicz [11], Paul [13] and Craig, Hsu and Sastry [3], have proposed the use of a computed torque controller for robots. This method, which involves computation and cancellation of all non-linearities and cross-coupling terms, shall now be described. Given some reference signal $r(t)$ our objective shall be to make θ track θ^*, where

$$\left(D^2 + \beta_1 D + \beta_2\right)\theta^* = \beta_2 r \tag{19}$$

and β_1 and β_2 are positive scalars. In the case where all the parameters are known, the following control achieves this objective (for any positive constants, γ_1 and γ_2):

$$\tau_M = -\tau_G(\theta) - \frac{1}{2}\frac{\partial}{\partial\theta}\left[\dot{\theta}^{\mathrm{T}}I(\theta)\dot{\theta}\right] + \dot{I}(\theta)\dot{\theta} - I(\theta)\left[\gamma_1\dot{\theta} + \gamma_2\theta - \ddot{\theta}^* - \gamma_1\dot{\theta}^* - \gamma_2\theta^*\right], \tag{20}$$

since upon substitution of (20) into (1) we obtain

$$I(\theta)[\ddot{\varepsilon} + \gamma_1\dot{\varepsilon} + \gamma_2\varepsilon] = 0 \tag{21}$$

where

$$\varepsilon \triangleq \theta - \theta^*. \tag{22}$$

In the case where the parameters are not known but are estimated as in Section 3, we propose the following adaptive computed torque control law:

$$\tau_M = T_1 + T_2 + T_3 + T_4 \tag{23}$$

where

$$T_1 = -G(\theta) - D(\theta, \dot{\theta}) + \dot{J}(\theta)\dot{\theta} + \sum_{k=1}^{N} \left(\dot{K}_k(\theta)\dot{\theta} - E_k(\theta, \dot{\theta}) - H_k(\theta)\right)\hat{m}_k, \tag{24}$$

$$T_2 = -\hat{I}(\theta)\left[\gamma_1(\dot{\theta} - \dot{\theta}^*) + \gamma_2(\theta - \theta^*) - \ddot{\theta}^*\right], \tag{25}$$

$$T_3 = \frac{1}{w}\Psi^{\mathrm{T}}\dot{\hat{M}}, \tag{26}$$

$$T_4 = -\frac{1}{w}\hat{I}(\theta)\left[\frac{\mathrm{d}}{\mathrm{d}t}\{\hat{I}^{-1}(\theta)\}\right]e, \tag{27}$$

and

$$\hat{I}(\theta) = J(\theta) + \sum_{k=1}^{N} K_k(\theta)\hat{m}_k. \tag{28}$$

The terms T_1 and T_2 in (23) are the certainty equivalence adaptive form of the computed torque controller (20). T_3 and T_4 have been included to deal with terms which arise in the adaptive case due to the commuting of time varying operators.

4. Global convergence for adaptive computed torque control

In this section we shall establish global convergence for the adaptive computed torque controller. To achieve this, we shall make use of the following result.

Lemma 2. *Suppose H is a linear, strictly proper, exponentially stable transfer function, and f_1 belongs to L_1, f_2 belongs to L_2 and f_∞ is bounded. Then $g = H^*(f_1 + f_2 + f_\infty)$ is bounded.*

The proof is given in the appendix.

Technical difficulties arise in the global convergence proof if $\hat{I}(\theta)$ is not uniformly positive definite. For this reason we shall assume the knowledge of a convex region, $C \subset \mathbb{R}^N$ such that $M \in C$ and $\forall\ \hat{M} \in C$, $\forall\ \theta \in \mathbb{R}^n$, $\hat{I}(\theta)$ is uniformly positive definite, i.e. $\hat{I}^{-1}$ is bounded. In this case it is a simple task to alter the parameter estimator (15) and (16) such that $\hat{M}$ belongs to C for all time, and properties (i) and (ii) of Lemma 1 hold. (See [6] for details of this modification.) The following lemma will also be used in the global convergence proof.

Lemma 3. *Suppose H is a linear, strictly proper exponentially stable transfer function, and $V(t)$ satisfies*

$$V(t) = H^*(V(t)[f_1 + f_2] + g_1 + g_2 + g_\infty) \tag{29}$$

and

$$V(t) = V_\infty + V_2, \tag{30}$$

where f_1, f_2, g_1, g_2, g_∞, V_2 and V_∞ are functions of time and f_1, g_1 belong to L_1; f_2, g_2 and V_2 belong to L_2 and g_∞, V_∞ are bounded. Then $V(t)$ is bounded.

For the proof, see the appendix. We now proceed to our main result.

Theorem 1. *Consider the adaptive computed torque control law* (23) *applied to the system* (1) *with parameter estimator* (15) *and* (16), *modified to ensure* $\hat{I}^{-1}(\theta)$ *is bounded. Then provided* $\boldsymbol{r}(t)$ *is bounded, this adaptive controller is globally convergent, i.e.*

(i) $\boldsymbol{\theta}$, $\dot{\boldsymbol{\theta}}$ *and* $\ddot{\boldsymbol{\theta}}$ *are bounded.*

(ii) $\boldsymbol{e}$, $\boldsymbol{\varepsilon}$ *and* $\dot{\boldsymbol{\varepsilon}}$ *tend to zero* (*where* $\boldsymbol{\varepsilon}$ *is defined in* (22)).

Proof. (i) Operating on the left of equation (13) by $(D+w)/w$ we obtain

$$\left(\frac{D+w}{w}\right)\boldsymbol{e} = \boldsymbol{Y} + \boldsymbol{\tau}_M - \boldsymbol{\phi}^{\mathrm{T}}(\hat{\boldsymbol{M}}) - \frac{1}{w}\boldsymbol{\Psi}^{\mathrm{T}}\dot{\hat{\boldsymbol{M}}}. \tag{31}$$

(The reason for $\boldsymbol{T}_3$ in (23) thus becomes clear.) Also from (24) and (6) we can show that

$$\boldsymbol{Y} + \boldsymbol{T}_1 - \boldsymbol{\phi}^{\mathrm{T}}(\hat{\boldsymbol{M}}) = -\hat{I}(\boldsymbol{\theta})\ddot{\boldsymbol{\theta}}. \tag{32}$$

Thus using (32), (23) to (27) in (31) we see that

$$-\left(\frac{D+w}{w}\right)\boldsymbol{e} = \hat{I}(\boldsymbol{\theta})(\ddot{\boldsymbol{\varepsilon}} + \gamma_1\dot{\boldsymbol{\varepsilon}} + \gamma_2\boldsymbol{\varepsilon}) + \frac{1}{w}\hat{I}(\boldsymbol{\theta})\left[\frac{\mathrm{d}}{\mathrm{d}t}\{\hat{I}^{-1}(\boldsymbol{\theta})\}\right]\boldsymbol{e} \tag{33}$$

or

$$-\left(\frac{D+w}{w}\right)\left[\hat{I}^{-1}(\boldsymbol{\theta})\boldsymbol{e}\right] = \ddot{\boldsymbol{\varepsilon}} + \gamma_1\dot{\boldsymbol{\varepsilon}} + \gamma_2\boldsymbol{\varepsilon} \tag{34}$$

and thus we see the reason for introducing $\boldsymbol{T}_2$ in (23). We define $\boldsymbol{f} = -(1/w)\hat{I}^{-1}\boldsymbol{e}$ and note that since $\hat{I}^{-1}$ is bounded and $\boldsymbol{e}$ belongs to L_2, then $\boldsymbol{f}$ belongs to L_2. We then have

$$\left[D^2 + \gamma_1 D + \gamma_2\right]\boldsymbol{\varepsilon} = [D+w]\boldsymbol{f} \tag{35}$$

and so $\boldsymbol{\varepsilon}$ is bounded, $\boldsymbol{\varepsilon}$ and $\dot{\boldsymbol{\varepsilon}}$ belong to L_2. The remainder of the proof are the technicalities required to ensure that we have boundedness (i.e. L_∞) as well as L_2 properties, and that we have uniform continuity of $\boldsymbol{\epsilon}$ and $\dot{\boldsymbol{\epsilon}}$, so that convergence can be assured. To establish boundedness of $\dot{\boldsymbol{\theta}}$ and $\ddot{\boldsymbol{\theta}}$, we shall use the torque expression, (23), and the robot dynamical equation (1). $\boldsymbol{T}_3$ of (26) can be rewritten as

$$\boldsymbol{T}_3 = -\frac{\alpha}{w}(\boldsymbol{\Psi}^{\mathrm{T}}P\boldsymbol{\Psi})\boldsymbol{e}. \tag{36}$$

P and w are bounded, $\boldsymbol{e}$ belongs to L_2 and so we shall consider $\boldsymbol{\Psi}$. Using (8), we can show that the kth row of $\boldsymbol{\Psi}$ is

$$\boldsymbol{\Psi}_k = \left[wK_k(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} - \left(\frac{w}{D+w}\right)\{wK_k(\boldsymbol{\theta})\dot{\boldsymbol{\theta}} + \boldsymbol{H}_k(\boldsymbol{\theta}) + \boldsymbol{E}_k(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})\}\right]. \tag{37}$$

Since $\boldsymbol{\theta}$ is bounded, we have that $K_k(\boldsymbol{\theta})$ and $H_k(\boldsymbol{\theta})$ are both bounded. Also, from the definition of $\boldsymbol{E}_k(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})$, we note that since $\boldsymbol{\theta}$ is bounded, $\boldsymbol{E}_k$ is bounded by $\dot{\boldsymbol{\theta}}^2$, i.e. $\exists\ c_E < \infty$, such that $\|\boldsymbol{E}_k(\boldsymbol{\theta}, \dot{\boldsymbol{\theta}})\| \leqslant c_E\|\dot{\boldsymbol{\theta}}\|^2$ for all $\dot{\boldsymbol{\theta}}$, $\boldsymbol{\theta}$.

Since $\dot{\boldsymbol{\varepsilon}} \in L_2$,

$$\dot{\boldsymbol{\theta}} = (L_2) + (L_\infty) \tag{38}$$

(where '(L_p)' denotes some function belonging to the space L_p). We then have from (37) and Lemma 2 that

$$\boldsymbol{\Psi} = (L_\infty)\dot{\boldsymbol{\theta}} + (L_\infty) = (L_\infty) + (L_2). \tag{39}$$

Now considering (36), we have

$$\boldsymbol{T}_3 = [(L_2) + (L_1)]\dot{\boldsymbol{\theta}} + (L_2) + (L_1) \tag{40}$$

(since P is bounded and $\boldsymbol{e}$ belongs to L_2).

We shall now consider T_4 in more detail:

$$T_4 = \frac{1}{w}\left[J(\dot{\theta}) + \sum_{k=1}^{N}\left(\dot{K}_k\hat{m}_k + K_k\dot{\hat{m}}_k\right)\right]\hat{I}^{-1}(\theta)e. \tag{41}$$

Since θ is bounded,

$$\dot{J}(\theta) = \frac{\partial J(\theta)}{\partial\theta}\dot{\theta} = (L_\infty) + (L_2) \tag{42}$$

and

$$\dot{K}_k(\theta) = (L_\infty) + (L_2), \tag{43}$$

$$\dot{\hat{M}} = P\Psi e = \left[(L_\infty)\dot{\theta} + (L_\infty)\right](L_2) \quad \text{(using (39))}$$

$$= (L_2)\dot{\theta} + (L_2). \tag{44}$$

Using (42)--(44) in (41), $e \in L_2$ and $\hat{I}^{-1}$ is bounded,

$$T_4 = (L_1)\dot{\theta} + (L_2) + (L_1). \tag{45}$$

We then have

$$\tau_m = \left[(L_1) + (L_2)\right]\dot{\theta} + (L_1) + (L_2) + (L_\infty). \tag{46}$$

From (1), (38), and, using the fact that θ is bounded,

$$\ddot{\theta} = \left[(L_1) + (L_2)\right]\dot{\theta} + (L_1) + (L_2) + (L_\infty). \tag{47}$$

Adding (38) to (47) and rearranging gives

$$\dot{\theta} = \frac{1}{1+D}\left\{\left[(L_1) + (L_2)\right]\dot{\theta} + (L_1) + (L_2) + (L_\infty)\right. . \tag{48}$$

Equations (48) and (38) in Lemma 3 now give that $\dot{\theta}$ is bounded. Then arguing as before (i.e. via the torque (i.e. via the torque expression (23) and then the robot dynamic equation (1)), τ_m is bounded and $\ddot{\theta}$ is bounded.

(ii) From (14) it is clear that

$$\dot{e} = \dot{\Psi}^{\mathrm{T}}\hat{M} + \Psi^{\mathrm{T}}\dot{\hat{M}}. \tag{49}$$

In view of (i), Ψ and $\dot{\Psi}$ are bounded. Then using (13), e is bounded and so $\dot{\tilde{M}} = \dot{\hat{M}}$ is bounded using (15). Thus it is clear that e is uniformly continuous. Since e belongs to L_2, we then have $e \to 0$. From the definition of f and (35), it is clear that f, ε and $\dot{\varepsilon} \to 0$. □

5. Conclusion

In this paper, we have considered the control of rigid link mechanical manipulators. We have then demonstrated global convergence for an adaptive computed torque control law which requires measurements of state variables only. Further work on robustness, computational aspects and performance considerations warrants further research.

Appendix

Proof of Lemma 2. Let $h(t)$ denote the impulse response of H. Then,

$$g(t) = \int_0^t h(t-\tau)[f_1(\tau) + f_2(\tau) + f_\infty(\tau)]\,\mathrm{d}\tau, \tag{A.1}$$

$$\|g(t)\| \leqslant \int_0^t \|h(t-\tau)\|[\|f_1(\tau)\| + \|f_2(\tau)\| + \|f_\infty(\tau)\|]\,\mathrm{d}\tau. \tag{A.2}$$

In view of the assumptions regarding H, ∞ c, $\sigma > 0$ such that

$$\|h(t-\tau)\| \leqslant c\,\mathrm{e}^{-\sigma(t-\tau)}. \tag{A.3}$$

Using (A.3) and Schwarz's inequality in (A.2) gives

$$\|g(t)\| \leqslant \left[\int_0^t c^2\,\mathrm{e}^{-2\sigma(t-\tau)}\,\mathrm{d}\tau\right]^{1/2}\left[\int_0^t \|f_2(\tau)\|^2\,\mathrm{d}\tau\right]^{1/2} + c\int_0^t \|f_1(\tau)\|\,\mathrm{d}\tau + \left[\sup_{0\leqslant\tau\leqslant t}\{\|f_\infty(\tau)\|\}\right]\int_0^t c\,\mathrm{e}^{-\sigma(t-\tau)}\,\mathrm{d}\tau. \tag{A.4}$$

From (A.4), and using the assumptions concerning f_1, f_2 and f_∞ it is clear that g is bounded. □

Proof of Lemma 3. From Lemma 2, $H^*(g_1 + g_2 + g)$ is bounded. Therefore we need only consider $H^*(V(t)[f_1 + f_2])$. If we let $h(t)$ denote the impulse response of H, then the response due to $V(t)[f_1 + f_2]$ is

$$V(t) = \int_0^t h(t-\tau)V(\tau)(f_1(\tau) + f_2(\tau))\,\mathrm{d}\tau. \tag{A.5}$$

Because of the assumptions on H, it follows that

$$\|V(t)\| \leqslant \int_0^t m\,\mathrm{e}^{-\lambda(t-\tau)}\|V(\tau)\|[|f_1(\tau)| + |f_2(\tau)|]\,\mathrm{d}\tau. \tag{A.6}$$

Using Schwarz's inequality we have

$$\|V(t)\| \leqslant \int_0^t m\|V(\tau)\||f_1(\tau)|\,\mathrm{d}\tau + m\left[\int_0^t \mathrm{e}^{-2\lambda(t-\tau)}\|V(\tau)\|\,\mathrm{d}\tau\right]^{1/2}\left[\int_0^t \|V(\tau)\|f_2^2(\tau)\,\mathrm{d}\tau\right]^{1/2}. \tag{A.7}$$

From Lemma 2 it follows that there exists a constant c such that

$$\|V(t)\| \leqslant \int_0^t m\|V(\tau)\|\|f_1(\tau)\|\,\mathrm{d}\tau + c\left[\int_0^t \|V(\tau)\|\|f_2(\tau)\|^2\,\mathrm{d}\tau\right]^{1/2} \tag{A.8}$$

$$\leqslant 1 + \int_0^t \|V(\tau)\|\|\bar{f}_1(\tau)\|\,\mathrm{d}\tau \tag{A.9}$$

where $\bar{f}_1(\tau) = m\|f_1(\tau)\| + c\|f_2(\tau)\|^2$ and thus $\bar{f}_1$ belongs to L_1. It can now be shown using Gronwall's lemma, that $V(t)$ is bounded. □

References

[1] G. Bastin, G. Campion and A.-M. Guillaume, Adaptive external linearization feedback control for manipulators, Technical Report, Université Catholique de Louvain, Louvain-la-Neuve, Belgium (1986).

[2] H. An Chae, C.G. Atkeson and J.M. Hollerback, Estimation of inertial parameters of rigid body links of manipulators, *24th CDC* (1985).

[3] J.J. Craig, P. Hsu and S.s. Sastry, Adaptive control of mechanical manipulators, Tech. Report, *Dept. of Elec. Eng. and Comp. Science,* University of California, Berkeley, CA (1985).

[4] S. Dubowsky and D.T. DesForges, The application of model reference adaptive control to robotic manipulators, *J. Dynamic Systems Measurement and Control* **101** (1979).

[5] H. Elliot, T. Depkovich, J. Kelly and B. Draper, Non-linear adaptive control of mechanical linkage systems with application to robotics, *Proc. of the American Control Conference*, San Diego, CA (1984).

[6] G.C. Goodwin and D.Q. Mayne, A parameter estimation perspective of continuous time adaptive control, *Automatica* **23** (1987) 57–70.

[7] P.K. Khasla and T. Kanade, Parameter identification of robot arms, *24th CDC* (1985).

[8] T.W.B. Kibble, *Classical Mechanics* (McGraw Hill, London, 1966).

[9] A.J. Koivo and T.H. Guo, Adaptive linear controller for robotic manipulators, *IEEE Trans. Automat. Control* **28** (2) 162–170.

[10] H.N. Koivo and J. Sorvari, On line tuning of a multivariable PID Controller for Robot Manipulators, *Proc. 24th CDC* (1985) 1502–1504.

[11] B. Markewicz, Analysis of a computed torque drive method and comparison with conventional position servo for a computer controlled manipulator, Tech. Memorandum 33-601, Jet Propulsion Lab., Pasadena, CA (1973).

[12] S. Nicosia and P. Tomei, Model reference adaptive control algorithms for industrial robots, *Automatica* **20** (5) (1984) 635–644.

[13] R.P. Paul, *Robot Manipulators: Mathematics, Programming and Control* (MIT Press, Cambridge, MA, 1981).

[14] M. Sahba and D.Q. Mayne, Computer aided design of non-linear controllers for torque controlled robot arms, *Proc. IEE* **131** (1) (1984) 8–14.

[15] S. Sastry and J.J. Slotine, Tracking control of non-linear systems using sliding surfaces and applications to robotics, *Internat. J. Comput.* (1983) 465–492.

[16] S.H. Singh, Adaptive model following control of nonlinear robotic systems, *IEEE Trans Automat. Control* **30** (11) 1099–1100.

[17] M.W. Spong and M. Vidyasagar, Robust linear compensator design for nonlinear robotic control, *Proc. IEEE Internat. Conf. Robotic Automation*, St. Louis, MO (March 1985).

[18] M. Tomizuka and R. Horowitz, Model reference adaptive control of mechanical manipulators, *IFAC Adaptive Systems in Control and Signal Processing*, San Francisco, CA (1983).